

PIL'VO, Russian Federation

WMO: 320690

Lat: 50.0462N

Lon: 142.1735E

Elev: 13

StdP: 14.69

Time Zone: 11.00 (E11)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.5	-12.1	-19.6	1.9	-14.7	-16.1	2.3	-11.4	35.5	15.0	31.6	14.5	10.5	120	0.726

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.9	71.7	65.2	69.7	64.3	67.7	63.1	66.9	70.1	65.4	68.4	63.9	66.9	10.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.6	94.6	69.0	64.0	89.4	67.5	62.5	84.7	66.2	31.6	70.3	30.4	68.5	29.3	66.9	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
30.3	26.0	21.2	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-21.0	76.0	4.9	2.4	-24.5	77.7	-27.3	79.1	-30.1	80.4	-33.6	82.2
			-21.2	69.2	4.8	2.9	-24.6	71.3	-27.4	73.0	-30.1	74.7	-33.6	76.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	34.2	5.3	6.7	18.0	31.8	42.2	51.3	59.3	61.5	54.2	41.2	25.8	11.4	(6)
(7)		DBStd	20.80	9.41	8.81	8.19	6.04	5.47	4.77	4.13	4.22	5.59	6.02	8.20	9.22	(7)
(8)		HDD50	6655	1384	1214	993	546	251	41	1	0	24	279	726	1196	(8)
(9)		HDD65	11263	1849	1634	1458	996	708	412	182	124	325	738	1176	1661	(9)
(10)		CDD50	888	0	0	0	0	9	79	288	357	149	6	0	0	(10)
(11)		CDD65	21	0	0	0	0	0	0	4	16	1	0	0	0	(11)
(12)		CDH74	15	0	0	0	0	0	0	1	14	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	9.3	10.3	9.9	9.5	9.2	8.4	7.7	7.0	7.6	9.3	10.4	11.4	11.1	(14)
(15)	Precipitation	PrecAvg	27.6	1.4	0.9	1.3	1.7	2.1	1.9	2.4	3.4	4.3	3.7	2.5	1.9	(15)
(16)		PrecMax	37.6	3.9	2.3	2.7	4.2	4.8	5.2	6.5	8.7	7.1	7.7	5.1	3.5	(16)
(17)		PrecMin	19.4	0.3	0.1	0.1	0.7	0.6	0.3	0.0	0.3	1.1	0.8	0.3	0.9	(17)
(18)		PrecStd	3.7	0.7	0.6	0.7	0.8	1.1	1.1	1.3	1.6	1.6	1.9	1.0	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.0	34.0	39.5	54.3	62.8	68.2	72.7	75.8	71.0	61.4	49.6	36.0	(19)
(20)		MCWB	29.9	31.2	34.3	43.2	51.3	57.7	64.6	68.6	62.9	54.3	44.8	33.7	(20)	
(21)		2%	DB	28.5	28.5	35.9	47.6	58.1	64.4	70.0	73.2	68.3	57.7	44.3	32.9	(21)
(22)		MCWB	26.8	26.7	32.1	40.2	49.8	56.8	64.2	66.9	62.1	51.9	40.5	31.0	(22)	
(23)		5%	DB	24.0	25.0	33.2	43.5	54.3	61.5	67.8	71.0	66.3	54.7	40.1	29.4	(23)
(24)		MCWB	22.5	23.1	30.2	38.3	47.6	55.9	63.0	65.4	61.0	49.7	36.8	27.4	(24)	
(25)		10%	DB	20.0	21.5	30.9	40.4	51.3	58.9	65.9	69.1	64.3	51.9	36.8	25.5	(25)
(26)		MCWB	18.8	20.1	28.4	36.2	46.2	54.5	62.0	64.6	59.4	47.3	34.1	23.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.4	31.3	35.2	45.2	54.5	60.8	66.7	70.7	65.7	56.4	45.7	34.1	(27)
(28)		MDCB	32.4	34.0	38.0	51.8	59.9	65.2	70.2	74.4	69.0	60.0	48.6	35.7	(28)	
(29)		2%	WB	26.8	27.0	32.5	41.2	50.8	58.4	65.3	68.3	63.5	52.9	40.8	31.1	(29)
(30)		MDCB	28.9	28.5	35.0	46.6	56.1	62.4	68.5	71.6	66.8	56.7	43.7	32.7	(30)	
(31)		5%	WB	22.6	23.3	30.6	38.6	48.7	56.6	63.9	66.6	61.8	50.3	37.2	27.5	(31)
(32)		MDCB	23.9	24.8	33.0	42.9	53.3	60.3	66.7	69.6	65.3	53.9	39.9	29.2	(32)	
(33)		10%	WB	18.9	20.1	28.5	36.6	46.7	54.9	62.5	65.3	60.0	47.8	34.2	23.8	(33)
(34)		MDCB	20.1	21.6	30.7	39.9	50.4	58.1	65.2	68.2	63.8	51.5	36.2	25.5	(34)	
(35)	Mean Daily Temperature Range		MDBR	15.3	17.1	15.5	11.0	11.0	10.1	9.0	10.9	14.2	12.6	10.5	12.5	(35)
(36)		5% DB	MCDBR	13.6	16.7	15.8	16.8	17.4	15.4	11.7	13.1	15.8	17.2	13.7	13.2	(36)
(37)			MCWBR	12.5	15.3	13.2	10.8	10.6	9.3	7.2	8.6	11.4	13.3	11.3	12.1	(37)
(38)			5% WB	MCDBR	13.4	15.6	14.9	15.1	14.8	12.7	9.4	10.3	13.5	16.3	13.6	12.8
(39)		MCWBR		12.5	14.5	12.9	10.6	9.9	8.6	6.5	7.7	10.6	13.6	11.6	12.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.298	0.333	0.406	0.507	0.548	0.506	0.476	0.442	0.396	0.412	0.371	0.311	(40)	
(41)		taud	2.047	1.942	1.817	1.762	1.801	2.011	2.170	2.284	2.340	2.096	2.044	2.049	(41)	
(42)		Ebn at Noon	222	243	243	232	230	242	249	252	252	213	188	194	(42)	
(43)		Edh at Noon	28	40	54	64	65	53	45	38	32	34	28	24	(43)	
(44)	All-Sky Solar Radiation	RadAvg	429	783	1163	1378	1535	1679	1503	1316	1094	678	380	301	(44)	
(45)		RadStd	39	50	87	141	147	160	172	119	95	53	39	36	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.63	+2.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	+0.61	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page